

DAKHLA, Egypt

WMO: 624320

Lat: 25.5028N

Lon: 28.9725E

Elev: 111

StdP: 100.00

Time Zone: 2.00 (E02)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	5.7	-3.7	2.8	13.3	-2.2	3.2	12.6	7.4	19.6	6.7	18.8	0.9	270	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	43.0	22.1	41.9	21.6	40.8	21.3	23.3	39.8	22.7	39.1	22.1	38.5	3.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.7	32.4	16.4	11.8	32.2	15.6	11.2	32.0	69.7	40.3	67.2	39.3	65.1	38.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.3	5.5	DB	2.0	44.8	1.3	1.0	45.6	0.3	46.2	-0.4	46.8	-1.4	47.6		
			WB	0.0	24.9	1.3	1.1	25.7	-1.7	26.3	-2.4	27.0	-3.3	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	14.4	16.1	20.0	24.5	28.7	31.4	32.3	32.5	30.5	26.2	20.2	15.5
	DBStd	7.12	2.37	3.14	3.40	3.63	3.20	2.25	1.75	1.78	2.30	3.00	2.79	2.25
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	324	123	78	19	1	0	0	0	0	0	12	91	
	CDD10.0	5259	138	170	311	434	579	642	692	696	616	504	304	172
	CDD18.3	2542	2	15	72	186	321	392	433	438	366	246	67	4
	CDH23.3	37153	80	258	976	2479	4656	6068	6938	6892	5192	2838	684	94
	CDH26.7	23133	14	83	423	1358	2918	4026	4693	4620	3257	1509	216	16
Wind	WSAvg	2.0	1.8	2.0	2.1	2.2	2.2	2.3	2.0	1.9	2.1	2.0	1.8	1.7
Precipitation	PrecAvg	1	0	0	0	0	0	0	0	0	0	0	0	0
	PrecMax	11	5	2	4	6	0	0	0	0	0	5	0	0
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	2	1	0	1	1	0	0	0	0	0	1	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	33.3	37.3	41.7	43.6	44.3	43.7	44.0	42.9	40.4	34.1	28.6
		MCWB	16.0	18.1	19.3	20.7	21.9	22.4	21.8	22.9	22.6	21.2	19.2	16.3
	2%	DB	25.4	29.0	34.1	38.9	41.4	42.4	42.2	42.4	41.1	37.6	31.2	25.8
		MCWB	14.4	15.9	17.8	19.9	21.2	21.6	21.5	22.4	21.8	20.7	18.0	15.3
	5%	DB	23.1	26.4	31.2	36.4	39.6	40.9	41.1	41.0	39.4	35.5	29.0	23.8
		MCWB	13.5	14.6	16.7	18.9	20.5	21.0	21.3	21.9	21.4	19.9	17.3	14.3
	10%	DB	21.4	23.9	28.6	33.9	37.6	39.2	40.0	39.8	37.8	33.6	27.2	22.2
		MCWB	12.8	13.5	15.3	17.9	19.8	20.6	21.1	21.5	20.8	19.2	16.6	13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.5	18.8	20.2	21.7	23.0	23.5	23.6	24.2	24.0	22.7	20.1	17.4
		MCDB	26.9	31.4	35.0	38.8	40.5	41.3	40.1	40.2	40.1	37.1	30.8	27.1
	2%	WB	14.9	16.5	18.4	20.3	22.0	22.4	22.6	23.4	22.8	21.4	18.7	15.6
		MCDB	24.3	28.1	32.5	37.3	39.4	40.1	39.5	39.8	38.0	34.9	29.4	24.6
	5%	WB	13.8	14.9	17.0	19.2	21.1	21.6	22.0	22.7	22.1	20.5	17.7	14.7
		MCDB	22.4	25.1	29.9	35.1	38.0	38.8	39.0	39.0	37.3	33.6	27.8	23.1
	10%	WB	12.9	13.8	15.8	18.1	20.1	21.0	21.5	22.1	21.4	19.7	16.8	13.8
		MCDB	21.0	23.5	27.7	33.1	36.3	37.8	38.3	38.3	36.2	32.4	26.5	21.7
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.6	16.0	16.8	16.4	15.6	15.4	15.0	14.6	14.2	14.4	14.4
		MCDBR	17.3	18.7	20.1	20.4	18.9	17.6	16.9	16.5	16.4	16.3	16.6	16.6
		MCWBR	9.6	9.3	8.8	8.3	7.3	6.1	6.2	6.1	5.9	6.5	7.8	9.0
	5% WB	MCDBR	16.4	18.1	18.7	19.4	18.5	16.7	15.9	15.3	15.7	15.3	15.7	15.7
		MCWBR	9.4	9.4	8.5	8.2	7.5	6.3	6.4	6.0	6.3	6.6	7.9	8.9
Clear-Sky Solar Irradiance	taub	0.388	0.431	0.486	0.587	0.601	0.523	0.507	0.500	0.491	0.478	0.409	0.391	
	taud	2.059	1.918	1.783	1.597	1.588	1.765	1.816	1.841	1.851	1.864	2.015	2.049	
	Ebn at Noon	847	837	812	742	731	784	796	801	797	780	816	824	
	Edh at Noon	146	181	219	270	272	226	215	209	202	189	151	142	
All-Sky Solar Radiation	RadAvg	4.52	5.42	6.51	7.31	7.79	8.12	7.91	7.50	6.78	5.73	4.78	4.24	
	RadStd	0.11	0.20	0.21	0.16	0.15	0.09	0.08	0.08	0.15	0.09	0.08	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.30	N/A	N/A	N/A	-0.95	-1.70	N/A	N/A	N/A	N/A

Nomenclature: See separate page